Symmetry Reduction for Sum-of-Squares Optimization

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January, 2022

INFORMS 2022

Nonnegative polynomial into sum of squares (SOS)

$$(x^3, x^2, x) \qquad \textit{not} \text{ unique}$$

$$p(x) = X^\top Q X$$

$$x^6 - 2x^4 + 2x^2 \; (^1) \quad = \quad X^\top \begin{pmatrix} 1 & 0 & -1 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 2 \end{pmatrix} X$$

$$p(x) \geq 0 \;\forall x \quad \Longleftarrow \quad Q \succeq 0 \qquad \Big\downarrow \text{cholesky}$$

$$x^2 + (-x^3 + x)^2 \quad \longleftarrow \quad X^\top \begin{pmatrix} 0 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix}^\top \begin{pmatrix} 0 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix} X$$

[1] Gatermann, Parrilo, *Symmetry groups, semidefinite programs, and sums of squares*, 2004: Example 5.1

Semidefinite program

$$\begin{aligned} \min_{X \in \mathcal{S}^N} \quad & \langle C, Q \rangle \\ \text{subject to:} \quad & \langle A_i, Q \rangle = b_i, \quad i = 1, 2, \ldots, m \\ & Q \succeq 0. \end{aligned}$$

Is $p(x) = x^6 - 2x^4 + 2x^2$ SOS[2] ? $p(x) = X^\top Q X = \langle X X^\top, Q \rangle$:

$$X = \begin{bmatrix} x^3 \\ x^2 \\ x \end{bmatrix}, XX^\top = \begin{bmatrix} x^6 & x^5 & x^4 \\ \cdot & x^4 & x^3 \\ \cdot & \cdot & x^2 \end{bmatrix}, A_{x^6} = \begin{bmatrix} 1 & 0 & 0 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 0 \end{bmatrix}, A_{x^5} = \begin{bmatrix} 0 & 1 & 0 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 0 \end{bmatrix},$$

$$A_{x^4} = \begin{bmatrix} 0 & 0 & 1 \\ \cdot & 1 & 0 \\ \cdot & \cdot & 0 \end{bmatrix}, A_{x^3} = \begin{bmatrix} 0 & 0 & 0 \\ \cdot & 0 & 1 \\ \cdot & \cdot & 0 \end{bmatrix}, A_{x^2} = \begin{bmatrix} 0 & 0 & 0 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 1 \end{bmatrix}.$$

[2] Gatermann, Parrilo, *Symmetry groups, semidefinite programs, and sums of squares*, 2004: Example 5.1

SDP solvers

For n-variate polynomial of degree $2d$: $N = \binom{n+d}{n}, m = \binom{n+2d}{n}$.

Complexity of second order interior point solvers[3]

Time: $\mathcal{O}((mN + m^2)N^{5/2}\log(\varepsilon))$. Space: $\mathcal{O}(m^2)$.

E.g. CSDP, DSDP, Hypatia, MOSEK, SDPA, SDPT3, SeDuMi.

Alternatives:

- Newton conjugate gradient (Newton-CG), e.g., SDPNAL.
- ADMM, e.g., CDCS, COSMO, SCS.
- Barvinok-Pataki bound: $\exists Q$ of rank $r \leq \sqrt{2m}$.
  SOSP: $\exists Q$ of rank $r \leq$ pythagoras number.
  - Burer-Monteiro method: $Q = UU^\top$, $U \in \mathbb{R}^{N \times r}$, e.g. SDPLR.
  - Primal-dual hybrid gradient (PDHG), e.g., ProxSDP.

[3] Monteiro, Todd. "Path-following methods." *Handbook of semidefinite programming*. 2000. 267-306.

Schur lemma

$(\rho_i)_i \subseteq \mathbb{R}^{d\times d}$ *reducible* if $\exists \{0\} \subset \mathcal{V} \subseteq \mathbb{R}^d$ s.t. $\forall i, \rho_i \mathcal{V} \subseteq \mathcal{V}$.

$(\rho_i^{(1)})_i, (\rho_i^{(2)})_i \subseteq \mathbb{R}^{d\times d}$ *equivalent* if $\exists T$ s.t. $\forall i, \rho_i^{(1)} = T\rho_i^{(2)}T^{-1}$.

If $(\rho_i^{(1)})_i, (\rho_i^{(2)})_i$ are irreducible and inequivalent, then

$$\forall i, Q\begin{bmatrix}\rho_i^{(1)} & 0\\ 0 & \rho_i^{(2)}\end{bmatrix} = \begin{bmatrix}\rho_i^{(1)} & 0\\ 0 & \rho_i^{(2)}\end{bmatrix} Q \Rightarrow \exists c_1, c_2 \text{ s.t. } Q = \begin{bmatrix}c_1 I & 0\\ 0 & c_2 I\end{bmatrix}$$

If $(\rho_i)_i$ is irreducible, then (permuting rows/columns give $I_d \otimes C$)

$$Q\begin{bmatrix}\rho_i & 0\\ 0 & \rho_i\end{bmatrix} = \begin{bmatrix}\rho_i & 0\\ 0 & \rho_i\end{bmatrix} Q \Rightarrow \exists C \in \mathcal{S}^2 \text{ s.t. } Q = C \otimes I_d = \begin{bmatrix}c_{1,1} I & c_{1,2} I\\ c_{1,2} I & c_{2,2} I\end{bmatrix}$$

$$\forall i, Q(\rho_i \otimes I_m) = (\rho_i \otimes I_m)Q \Rightarrow Q = C \otimes I_d, C \in \mathcal{S}^m \quad \text{Note } m \leftrightarrow d$$

Sagan, *The symmetric group*, Springer Science & Business Media, 2001: Definition 1.4.5, Definition 1.6.2, Example 1.7.2, Example 1.7.3 (adapted from $\mathbb{C}^{d\times d}$ to symmetric matrices $\mathcal{S}^n \subseteq \mathbb{R}^{d\times d}$)

Invariance and convexity

Theorem[4]

Objective and feasible sets are invariant and convex $\longrightarrow$ can restruct to invariant solutions.

Invariant SDP: Q feasible $\Rightarrow \rho_i^\top Q \rho_i$ feasible w/ same cost:

$\forall i,j, \rho_i A_j \rho_i^\top = A_j$ and $\rho_i C \rho_i^\top = C$.

Invariant SOSP: $p(g_i(x)) = p(x)$, let ρ_i s.t. $X(g_i(x)) = \rho_i X(x)$.

Invariant solution: $Q = \rho_i^\top Q \rho_i$. If $\rho_i^\top = \rho_i^{-1}$, $\rho_i Q = Q \rho_i$.

How to simultaneously block-diagonalize $(\rho_i)_i$?

[4] Gatermann, Parrilo, *Symmetry groups, semidefinite programs, and sums of squares*, 2004: Theorem 3.3

Group Theory to the rescue!

Definition

A group $(G, *)$ is a set closed under the multiplication $*$, together with the existence of unique inverse elements.

Example: The set of all bijections of the set $\{1, \dots n\}$ with composition is a group (known as the full symmetric group on n letters S_n).

$S_2 = \{(1)(2), (1,2)\}$ acts on polynomial ring $\mathbb{R}[x]$ by

$$((1)(2), x) \mapsto x$$
$$((1,2), x) \mapsto -x.$$

Observe: $f(x) = x^6 - 2x^4 + 2x^2 = x^2 + (x - x^3)^2$ is invariant, but $x - x^3$ isn't!

Isotypical projections

"Types" of actions $\leftrightarrow$ blocks in the block-diagonalization:

$$V \cong V_1 \oplus \cdots \oplus V_k.$$

Each block V_i decomposes further as $V = \bigoplus_i \oplus_{m_i} V_i'$. ([5])

Universal (baseless) projections[6] in group ring $\mathbb{R}[G] \leftrightarrow$ projections onto blocks (for every representation V!).

Example

Universal **projections** for S_2:

$$\pi_1 = \frac{1}{2}((1)(2) + (1,2)) \to \text{invariants}$$

$$\pi_2 = \frac{1}{2}((1)(2) - (1,2)) \to \text{flipped}$$

[5] m_i is the multiplicity of the action type in V

[6] related to irreducible characters

Permutation symmetry example

```
using PermutationGroups
G = PermGroup([perm"(1,2,3,4)"])
# permutation group generated by one element (cyclic of order 4)
using DynamicPolynomials
@polyvar x[1:4]
poly = sum(x) + sum(x.^2)
basis = monomials(x, 0:(maxdegree(f)÷2))
using SymbolicWedderburn
symmetry_adapted_basis(G, basis, Symmetry.VariablePermutation())
# G acts by permuting variables
```

Isotypical blocks when acting on $[1, x_1, x_2, x_3, x_4, x_5]$:

$$B_1 = \begin{bmatrix} 1 \\ x_1 + x_2 + x_3 + x_4 \end{bmatrix} \qquad B_2 = \begin{bmatrix} x_1 - x_3 \\ x_2 - x_4 \end{bmatrix} \qquad B_3 = \begin{bmatrix} x_1 - x_2 + x_3 - x_4 \end{bmatrix}$$

$$V \cong (V_1' \oplus V_1') \oplus V_2' \oplus V_3', \qquad d_2 = m_1 = 2, \quad d_1 = d_3 = m_2 = m_3 = 1.$$

Permutation symmetry example

```
using SumOfSquares
import ECOS # Note: ECOS doesn't support PSD cone,
# but 2×2 psd constraints translate to SOC/quadratic constraints
model = Model(ECOS.Optimizer)
@variable(model, t)
@objective(model, Max, t)
pattern = Symmetry.Pattern(G, Symmetry.VariablePermutation())
@constraint(model, poly - t in SOSCone(), symmetry = pattern)
optimize!(model)
```

Naive: 15 vars 15 constrs, Q is 5×5; Symmetry: 5 vars 5 constrs,

$Q = Q_1 \oplus (Q_2 \oplus Q_2) \oplus Q_3 \qquad Q_1$ is $2 \times 2, Q_2$ is $1 \times 1, Q_3$ is 1×1

Note inversion: multiplicities $\leftrightarrow$ dimensions

$V \cong (V_1' \oplus V_1') \oplus V_2' \oplus V_3', \qquad d_2 = m_1 = 2, \quad d_1 = d_3 = m_2 = m_3 = 1.$

Even symmetry example[7]

```
struct OnSign <: Symmetry.OnMonomials end
# <: SymbolicWedderburn.Action, corresponds to x → -x
SymbolicWedderburn.action(::OnSign, p::Permutation,
    mono::AbstractMonomial) = sign(p)^degree(mono)*mono
G = PermGroup([perm"(1,2)"])
@polyvar x
f = x^6 - 2x^4 + 2x^2
basis = monomials(x, 0:(maxdegree(f)÷2))
symmetry_adapted_basis(G, basis, OnSign())
```

Isotypical blocks when acting on $[1, x, x^2, x^3]$:

$$B_1 = \begin{bmatrix} 1 \\ x^2 \end{bmatrix} \qquad B_2 = \begin{bmatrix} x \\ x^3 \end{bmatrix} \qquad V = (V_1' \oplus V_1') \oplus (V_2' \oplus V_2')$$

Naive formulation 4×4-psd; with symmetry: 2×2, 2×2-psd.

[7] Gatermann, Parrilo, *Symmetry groups, semidefinite programs, and sums of squares*, 2004: Example 5.1

Dihedral symmetry example[8]

```
@polyvar x y
f = x^6 + y^6 - x^4 * y^2 - y^4 * x^2 - x^4 - y^4 -
    x^2 - y^2 + 3x^2 * y^2 + 1
struct DihedralAction <:
    SymbolicWedderburn.ByLinearTransformation end
```

The action is a mixture of changing sign and permuting variables.

$$B_1 = [x^2 + y^2, 1], \qquad m_1 = 2, d_1 = 1,$$
$$B_2 = [x^3, x^2y, xy^2, y^3, x, y], \qquad m_2 = 3, d_2 = 2,$$
$$B_3 = [xy], \qquad m_3 = d_3 = 1,$$
$$B_4 = [x^2 - y^2], \qquad m_4 = d_4 = 1,$$

10×10-psd into 2×2, 3×3, 1×1, 1×1-psd.

Non-commutative groups allow $d_i > 1 \rightarrow$ better for the symmetrization!

[8] Gatermann, Parrilo, *Symmetry groups, semidefinite programs, and sums of squares*, 2004: Example 5.4

Implementing group action

- implement methods from `GroupsCore.jl` API for `DihedralGroup`

```
import GroupsCore
struct DihedralGroup <: GroupsCore.Group ... end
struct DihedralElement <: GroupsCore.GroupElement
    ...
end
```

- implement action methods from `SymbolicWedderburn` for `DihedralAction`

```
import SymbolicWedderburn
SymbolicWedderburn.action(
  ::DihedralAction,
  ::DihedralElement,
  ::AbstractMonomial) = ...
```

Behind the scenes: `SymbolicWedderburn.jl`

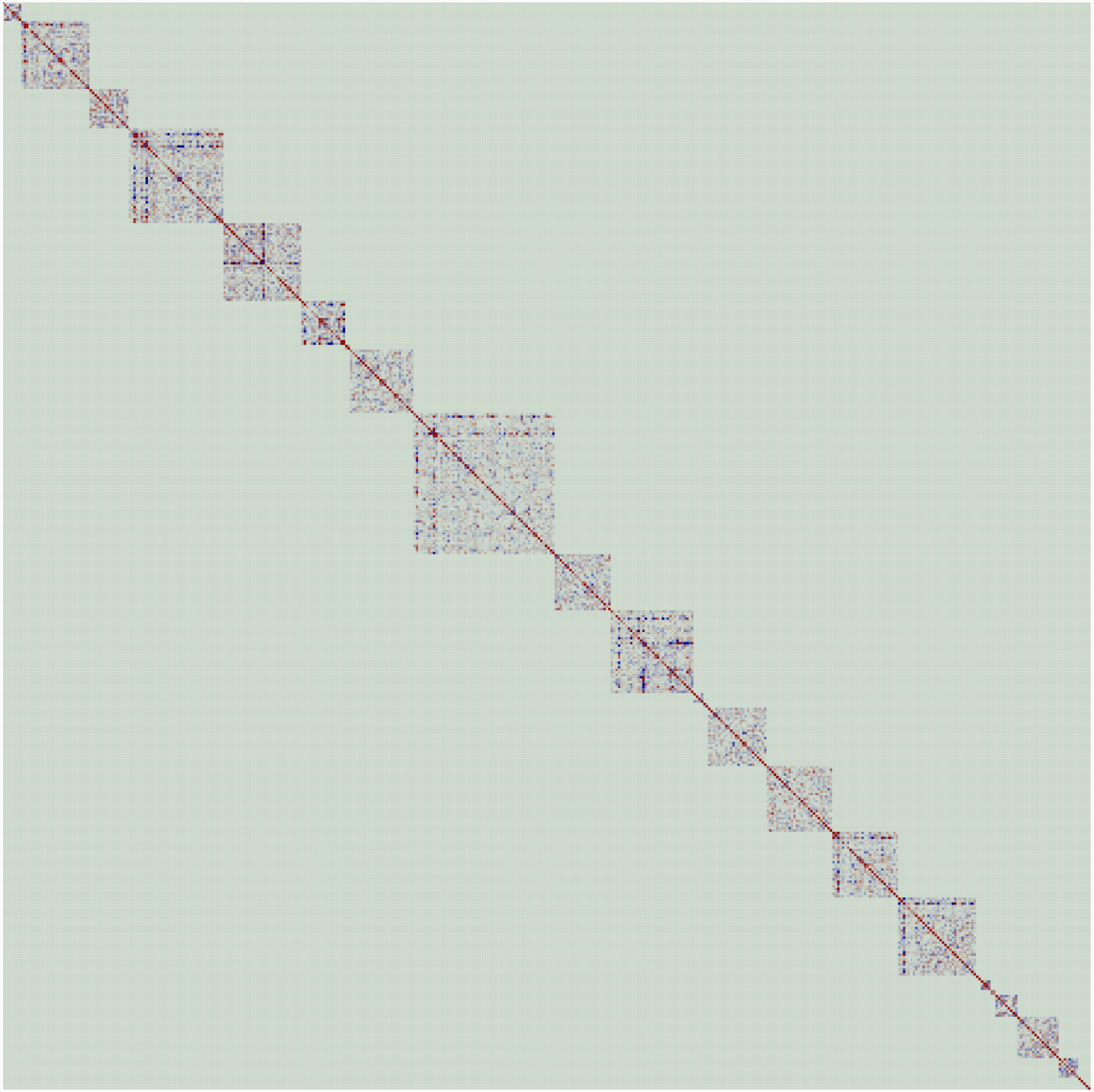
- compute character table of the group $\longrightarrow$ universal projections
- given action e.g. on monomials, induce the action to the whole basis
- evaluate the projections in the action $\longrightarrow$ block diagonalizing change of basis
- further decompose each block into irreducible blocks([9]).

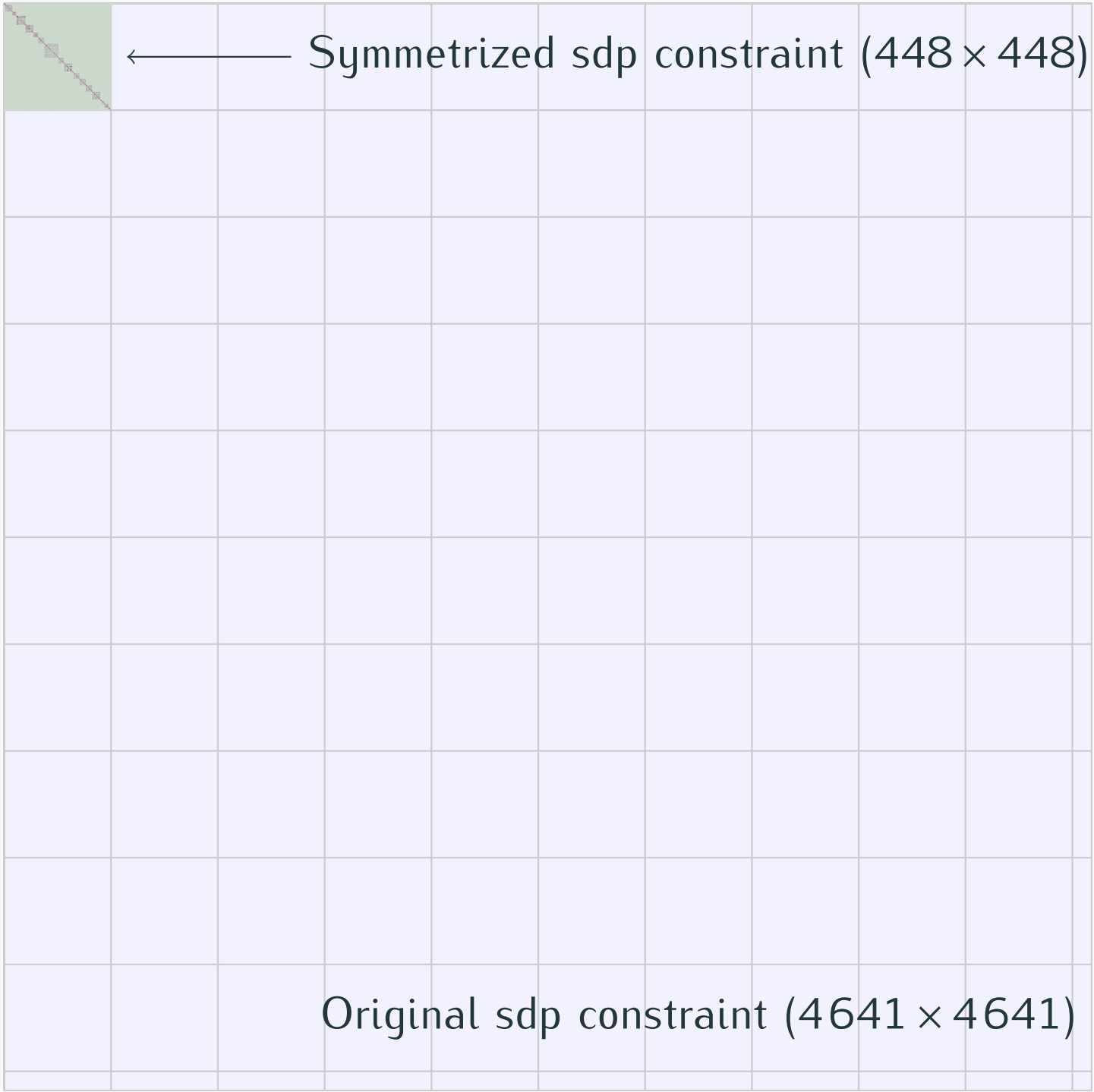
[9] for now numerically only

Large scale example

- Optimization problem from geometric group theory[10]
- sdp-constraint of size 4641×4641, $1.1 \cdot 10^7$ constraints
- symmetry group: $S_2 \wr S_5$ (3840 elements)
- After symmetrization:
  - 29-blocks (largest: 58×58), 13232 variables in total
  - 7230 constraints
- Solvable!

[10] Kaluba, M., Nowak, P.W. & Ozawa, N. Aut(F_5) has property (T). *Math. Ann.* **375**, 1169–1191 (2019). https://doi.org/10.1007/s00208-019-01874-9




← Symmetrized sdp constraint (448 × 448)
Original sdp constraint (4641 × 4641)